JEE Mathematics: Quick Review with Grading and Practice Map - Definite Integration (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (183 questions; a question that appears in two of these is counted once): High = 17 or more questions, Medium = 1 to 16 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction

Question 01

What does $\int_a^b f(x)\,dx$ represent for the graph of $y = f(x)$?

Model Answer

The algebraic sum of the areas between $y = f(x)$, the x-axis and $x = a$, $x = b$: area above the axis with $+$ sign, below with $-$ sign. So the value can be positive, zero or negative.

Ex: $\int_{-2}^{0} \sqrt{4 - x^2}\,dx = \pi$ (quarter circle of radius 2).

Grade - Medium

Practice map, main concept: Sheet: J-I 26; J-M 51

Also used in: none

Question 02

What to keep in mind while substituting in questions of definite integration?

Model Answer

Change the limits to the new variable (no need to come back to x). The substitution $t = \phi(x)$ must be continuous and monotonic (one-one) on the whole interval.

Ex: in $\int_0^{2\pi} \dfrac{dx}{2 + \sin x}$, $\tan\frac{x}{2} = t$ is not valid ($\tan\frac{x}{2}$ breaks at $x = \pi$); split at π first. Value $= \dfrac{2\pi}{\sqrt{3}}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 30, 32; J-II 41, 48, 50; J-M 28; J-A 6, 16, 27, 32, 41; C-I 6-7, 18; C-II 17-18 | DPP: D1(5-7, 9, 17, 20); D2(4, 11, 13, 23-24); D4(21); D5(9-10, 14-15) | Question Bank: 28, 46, 56, 81

Also used in: Sheet: J-I 57; J-II 77; J-A 8 | DPP: D2(3); D3(19) | Question Bank: 57

Question 03

What is the relation of the limits with dx in definite integration?

Model Answer

The limits are values of the variable written in the differential: in $\int_a^b f\,d(g(x))$ the limits are values of $g(x)$, not of x. After substitution, write the limits of the new variable.

Ex: $\int_{\ln x=0}^{\ln x=1} x\,d(\ln x) = \int_1^e x \cdot \dfrac{1}{x}\,dx = e - 1$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 04

What is the condition for the result $\int_a^b f'(x)\,dx = \big[f(x)\big]_a^b$?

Model Answer

$\int_a^b f'(x)\,dx = f(b) - f(a)$, valid if f is continuous on $[a, b]$

So any definite integral is found from an antiderivative (substitution, parts $\int_a^b uv' = \big[uv\big]_a^b - \int_a^b u'v$, partial fractions); check that the antiderivative has no break inside $[a, b]$, else split there.

Ex: $\int_0^\pi \sec^2 x\,dx \neq \big[\tan x\big]_0^\pi = 0$, because $\tan x$ breaks at $\frac{\pi}{2}$.

Grade - High

Practice map, main concept: Sheet: J-I 34-35, 45, 72; J-II 40, 42, 46-47, 49, 52, 111; J-M 3, 17, 54, 56, 70, 75, 79-80; J-A 5, 10, 13, 19, 34, 36; C-I 8-10, 19, 23; C-II 14-16, 19-20 | DPP: D1(1-4, 8, 11-16, 18-19, 21-25); D2(1-2, 5-10, 12, 14-15, 17-19, 21-22, 25, 28); D3(4); D5(1, 5, 18) | Question Bank: 21-22, 30, 35, 40, 42, 44, 47, 54-55, 57, 61, 64, 71, 88

Also used in: Sheet: J-II 43, 51, 61; J-M 7, 13, 28, 37-38, 46; J-A 2, 21, 25, 31; C-I 15; C-II 22-23 | DPP: D3(20, 28) | Question Bank: 49-50, 52, 72, 75, 82-83

Question 05

What is the condition for the result $\int_a^a f(x)\,dx = 0$? What follows when $\int_a^b f(x)\,dx = 0$?

Model Answer

$\int_a^a f(x)\,dx = 0$ if $x = a$ is in the domain of f. If f is continuous and $\int_a^b f(x)\,dx = 0$, then $f(x) = 0$ has at least one root in (a, b) (Rolle on $F(x) = \int_a^x f$).

Ex: $\int_0^1 (6x^2 - 6x + 1)\,dx = 0$, so $6x^2 - 6x + 1 = 0$ has a root in $(0, 1)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 27; J-A 26, 35 | DPP: D2(3)

Also used in: none

Question 06

What is the condition to bring the limit inside the integration sign?

Model Answer

$\lim_{n\to\infty} \int_p^q f_n(x)\,dx = \int_p^q \lim_{n\to\infty} f_n(x)\,dx$ if p, q are independent of n

Ex: $\lim_{n\to\infty} \int_0^1 \left(1 + \frac{x}{n}\right)^n dx = \int_0^1 e^x\,dx = e - 1$.

Grade - Medium

Practice map, main concept: Sheet: C-I 27 | DPP: D5(8) | Question Bank: 68

Also used in: none

Question 07

If $f(x)$ and $g(x)$ are inverses of each other and $f(a) = c$, $f(b) = d$, what is $\int_a^b f(x)\,dx + \int_c^d g(y)\,dy$?

Model Answer

$$\int_a^b f(x)\,dx + \int_c^d g(y)\,dy = bd - ac$$

Proof: put $y = f(x)$ in the second integral and use parts. Use it when one of the two integrals has no easy antiderivative.

Ex: $\int_1^e \ln x\,dx + \int_0^1 e^y\,dy = e \cdot 1 - 1 \cdot 0 = e$.

Grade - Medium

Practice map, main concept: Sheet: J-A 42 | DPP: D2(20, 27)

Also used in: Sheet: J-M 51

Question 08

What is the value of $\int_0^{\pi/2} \sin^n x\,dx$ and $\int_0^{\pi/2} \cos^n x\,dx$ if $n = 1, 2, 3, 4$?

Model Answer

Both are equal (king property): $1,\ \dfrac{\pi}{4},\ \dfrac{2}{3},\ \dfrac{3\pi}{16}$ for $n = 1, 2, 3, 4$.

For larger n use Wallis (38).

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 90

Question 09

What to substitute if the limits are reciprocals of each other?

Model Answer

Put $x = \dfrac{1}{t}$ (also when the limits are 0 and ∞), then add the new form to the old one.

Ex: $\int_{1/2}^{2} \dfrac{\ln x}{1+x^2}dx$: $x = \frac{1}{t}$ gives $I = -I$, so $I = 0$.

Grade - Medium

Practice map, main concept: Sheet: J-I 33; J-II 43-44, 51, 100; J-M 71; C-I 25 | DPP: D1(10); D5(17)

Also used in: Sheet: J-A 11 | Question Bank: 26

Property 2 of Definite Integral

Question 10

What will be the effect on the value of a definite integral if the limits are interchanged?

Model Answer

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

Only the sign changes. (P1: the variable of integration is a dummy, $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$.)

Grade - Low

Practice map, main concept: none

Also used in: none

Property 3 of Definite Integral

Question 11

Write property 3.

Model Answer

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

c need not lie between a and b.

Grade - Low

Practice map, main concept: DPP: D2(26)

Also used in: DPP: D2(12)

Question 12

In which type of definite integrals can P3 be used?

Model Answer

Non-uniform or discontinuous integrands: modulus, $[x]$, $\{x\}$, max/min, piecewise functions. Break the interval at the points where the formula changes (zeros inside $|\ |$, integer values inside $[\ ]$).

Ex: $\int_0^2 [x^2]\,dx = \int_1^{\sqrt{2}} 1\,dx + \int_{\sqrt{2}}^{\sqrt{3}} 2\,dx + \int_{\sqrt{3}}^{2} 3\,dx = 5 - \sqrt{2} - \sqrt{3}$.

Grade - High

Practice map, main concept: Sheet: J-I 36-39, 46; J-II 45, 53-55, 66, 103; J-M 2, 4, 9-10, 23-24, 29, 33, 44-45, 48-49, 53, 65, 67-68, 72, 77; J-A 12, 40, 46; C-I 11, 30; C-II 21 | DPP: D2(16); D3(1-3, 5); D5(7) | Question Bank: 23, 29, 37, 65-66, 69-70, 74, 78-79, 85

Also used in: Sheet: J-I 40, 42; J-II 71; J-M 59, 62; J-A 37; C-II 17, 39 | DPP: D3(7); D4(3, 7)

Question 13

If $f(y)$ is a non-negative continuous function such that $f(y) + f\left(y + \frac{1}{2}\right) = 7\ \forall\, y \geq 0$, find $\int_0^5 f(y)\,dy$.

Model Answer

Split $[0, 5]$ into pieces of length $\frac{1}{2}$ (P3). In $\int_{1/2}^{1} f(y)\,dy$ put $y = t + \frac{1}{2}$: it becomes $\int_0^{1/2} (7 - f(t))\,dt$. So each pair of neighbouring pieces gives $\frac{7}{2}$.

$$\int_0^5 f(y)\,dy = 5 \times \frac{7}{2} = \frac{35}{2}$$

Grade - Low

Practice map, main concept: DPP: D3(29)

Also used in: none

Property 4 of Definite Integral

Question 14

Write the property of definite integrals related to odd and even functions.

Model Answer

$$\int_{-a}^{a} f(x)\,dx = \int_0^a [f(x) + f(-x)]\,dx = \begin{cases} 2\int_0^a f(x)\,dx, & f \text{ even} \\ 0, & f \text{ odd} \end{cases}$$

Valid for discontinuous functions too. Split any integrand into even part + odd part; the odd part contributes 0.

Ex: $\int_{-\pi/2}^{\pi/2} \cos^2 x\,dx = 2\int_0^{\pi/2} \cos^2 x\,dx = \frac{\pi}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 40-43; J-II 56-59, 104; J-M 5, 11, 55; J-A 2, 38; C-I 13, 32; C-II 24 | DPP: D3(6-13); D4(7) | Question Bank: 25, 27, 50, 60, 76

Also used in: Sheet: J-II 72; J-M 4; J-A 6; C-I 14

Question 15

What is the pattern of limits when the odd and even property of definite integrals is used?

Model Answer

Limits $-a$ to a (opposite signs). If the limits are symmetric about c (e.g. $c-a$ to $c+a$), shift $x-c=t$ first. If f is neither odd nor even, use $\int_0^a [f(x)+f(-x)]dx$.

Ex: $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+3^x}dx = \int_0^{\pi} \cos^2 x\,dx = \frac{\pi}{2}$ (because $\frac{1}{1+3^x}+\frac{1}{1+3^{-x}}=1$).

Grade - Medium

Practice map, main concept: Sheet: J-I 50, 56; J-II 86; J-M 13, 25, 37, 41 | Question Bank: 33, 38, 83

Also used in: none

Property 5 of Definite Integral

Question 16

Write King property.

Model Answer

$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$; in particular $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Proof: put $x=a+b-t$. Valid for discontinuous functions too.

Ex: $\int_0^{\pi/2} \sin^2 x\,dx = \int_0^{\pi/2} \cos^2 x\,dx$; adding, $2I=\frac{\pi}{2}$, $I=\frac{\pi}{4}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 31, 48, 52; J-II 60-63, 69, 88, 106; J-A 21, 31, 37, 47; C-I 12, 14-15; C-II 23 | DPP: D3(14, 17, 19-20, 27-28); D4(1); D5(3, 11-12, 19) | Question Bank: 49, 72, 82

Also used in: Sheet: J-I 44, 74; J-II 66, 75, 102; J-M 17; C-II 16 | DPP: D3(26); D5(4) | Question Bank: 28

Question 17

What to do while applying King property on definite integrals?

Model Answer

Write I with x replaced by $a+b-x$ (limits unchanged), then add (or subtract) the two forms so that the integrand simplifies; after that, evaluate the simpler integral.

Ex: $I=\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x+\cos^3 x}dx$; king and add: $2I=\int_0^{\pi/2} dx$, $I=\frac{\pi}{4}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 18

In which type of definite integrals can King property be used?

Model Answer

$$\int_a^b \frac{f(x)}{f(x)+f(a+b-x)}dx = \frac{b-a}{2}$$

When the denominator stays the same after king (e.g. $\sin^n x+\cos^n x$, $\sqrt{x}+\sqrt{a+b-x}$, $1+\tan^k x$), and in $\ln(1+\tan x)$-type integrands.

Ex: $\int_2^4 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{6-x}}dx = \frac{4-2}{2}=1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 51, 55, 57; J-M 8, 15, 27, 39 | DPP: D3(16, 25) | Question Bank: 31

Also used in: none

Question 19
What is the condition on $f(x)$ when we apply king on $\int_a^b x\, f(x)\, dx$?

Model Answer

$$f(a+b-x) = f(x) \;\Rightarrow\; \int_a^b x\, f(x)\, dx = \frac{a+b}{2} \int_a^b f(x)\, dx$$

Typical: $\int_0^\pi x\, f(\sin x)\, dx = \frac{\pi}{2} \int_0^\pi f(\sin x)\, dx$.

Ex: $\int_0^\pi x \sin x\, dx = \frac{\pi}{2} \int_0^\pi \sin x\, dx = \pi$.

Grade - Medium

Practice map, main concept: Sheet: J-I 47, 49, 53-54; J-II 64-65, 67; C-II 40 | DPP: D3(15, 18, 21, 23-24) | Question Bank: 24, 48, 80

Also used in: Sheet: J-II 83 | Question Bank: 27

Property 6 of Definite Integral

Question 20
Write Queen property.

Model Answer

$$\int_0^{2a} f(x)\, dx = \int_0^a [f(x) + f(2a-x)]\, dx = \begin{cases} 2\int_0^a f(x)\, dx, & f(2a-x) = f(x) \\ 0, & f(2a-x) = -f(x) \end{cases}$$

Ex: $\int_0^\pi \sin^4 x\, dx = 2\int_0^{\pi/2} \sin^4 x\, dx = 2 \cdot \frac{3\pi}{16} = \frac{3\pi}{8}$.

Grade - Low

Practice map, main concept: Sheet: J-II 68, 89-90; C-II 22 | DPP: D3(22)

Also used in: Sheet: C-II 40

Question 21
What is the limitation while applying Queen property on definite integrals?

Model Answer
The lower limit must be 0 (if not, shift $x = t + a$ first or use king). It only reduces the upper limit to half.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 22
In which type of definite integrals can Queen property be used?

Model Answer
When $f(2a-x) = \pm f(x)$: powers and products of $\sin x$, $\cos x$ over $[0, \pi]$ or $[0, 2\pi]$, $|\sin x|$, $|\cos x|$; apply it repeatedly to reach $[0, \frac{\pi}{2}]$.

Ex: $\int_0^{2\pi} \sin^5 x \cos^4 x\, dx = 0$, because $f(2\pi - x) = -f(x)$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 23
How will we identify when to apply Queen property in definite integrals?

Model Answer
Lower limit 0, upper limit π, 2π (any $2a$) and the integrand built from $\sin x$, $\cos x$, so that we need to reduce the upper limit to $\frac{\pi}{2}$ and then use Wallis or king.

Ex: $\int_0^{\pi} |\cos x|\,dx = 2\int_0^{\pi/2} \cos x\,dx = 2$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 24
Write the result of $\int_0^{\pi/2} \ln(\sin x)\,dx$.

Model Answer

$$\int_0^{\pi/2} \ln(\sin x)\,dx = \int_0^{\pi/2} \ln(\cos x)\,dx = -\frac{\pi}{2}\ln 2$$

Proof: king, add, $\sin x \cos x = \frac{1}{2}\sin 2x$, then queen.

Ex: $\int_0^{\pi/4} \ln(1+\tan x)\,dx = \frac{\pi}{8}\ln 2$ (king gives $\ln \frac{2}{1+\tan x}$).

Grade - Low
Practice map, main concept: Sheet: J-I 44 | DPP: D3(26); D5(4)
Also used in: DPP: D3(24)

Property 7 of Definite Integral

Question 25
How to apply the property of periodic functions in definite integrals?

Model Answer
$f(x+T) = f(x)$: $\int_0^{nT} f = n\int_0^T f$, $\int_a^{a+nT} f = n\int_0^T f$, $\int_{mT}^{nT} f = (n-m)\int_0^T f$, $\int_{nT}^{a+nT} f = \int_0^a f$
Find the period ($|\sin x|$: π; $\{x\}$: 1), take out whole periods, and integrate the remaining part separately.

Ex: $\int_0^{16\pi/3} |\sin x|\,dx = 5\int_0^{\pi} |\sin x|\,dx + \int_0^{\pi/3} \sin x\,dx = 10 + \frac{1}{2} = \frac{21}{2}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 29, 58-60; J-II 70-71; J-M 7, 47, 59, 62, 64; J-A 15 | DPP: D4(2-6)
Also used in: Sheet: J-II 88

Differentiation of Integration

Question 26
How to differentiate an indefinite integral?

Model Answer

$$\frac{d}{dx}\int f(x)\,dx = f(x)$$

No need to integrate first; also $\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$ for constant a.

Grade - Low
Practice map, main concept: DPP: D5(6)

Also used in: none

Question 27

How to differentiate a definite integral (Leibnitz rule), and prove it.

Model Answer

$$\frac{d}{dx}\int_{g(x)}^{h(x)} f(t)\,dt = f(h(x))\,h'(x) - f(g(x))\,g'(x)$$

Proof: with $P'(t) = f(t)$, $\int_g^h f = P(h(x)) - P(g(x))$; differentiate by the chain rule (f continuous).

Ex: $F(x) = \int_x^{x^2} \ln t\,dt \Rightarrow F'(x) = 2x\ln x^2 - \ln x = (4x-1)\ln x$.

Grade - High

Practice map, main concept: Sheet: J-I 61-63, 76; J-II 73; J-M 14, 42-43, 46; J-A 4, 7-9, 11, 20, 24-25, 30, 39, 45; C-I 16, 26; C-II 32, 38 | DPP: D4(8-9); D5(2, 13, 16) | Question Bank: 34, 36, 59, 84

Also used in: Sheet: J-I 73; J-II 93; J-M 9-10, 16, 63, 69, 74, 78; J-A 34, 38, 44; C-I 30; C-II 34 | Question Bank: 62-63, 67

Question 28

What to keep in mind while applying Leibnitz rule?

Model Answer

The integrand must be free of x: take x outside (e.g. $\int_0^x (x-t)f(t)dt = x\int_0^x f - \int_0^x tf$) or substitute so that x goes into the limits. For $\frac{0}{0}$ limits containing $\int_0^{g(x)}$, use L'Hospital with Leibnitz.

Ex: $\lim\limits_{x\to 0} \dfrac{\int_0^{x^2} \sin\sqrt{t}\,dt}{x^3} = \lim\limits_{x\to 0} \dfrac{2x\sin x}{3x^2} = \dfrac{2}{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 64-70; J-II 72, 92; J-M 30; J-A 14; C-I 17; C-II 31 | Question Bank: 39, 45, 51, 53, 58, 77

Also used in: Sheet: J-A 35

Determination of Function

Question 29

How to solve questions on determination of a function if an identity involving a definite integral is given?

Model Answer

Constant limits: $\int_a^b f(t)\,dt$ is a constant k (or $A\sin x + B\cos x$ type after expanding); assume it, put back and solve for k. Variable limits: differentiate (Leibnitz) to get a differential equation; the starting value comes from putting the lower limit.

Ex: $f(x) = x + \int_0^1 tf(t)\,dt$: $f = x + k$, $k = \int_0^1 t(t+k)dt = \frac{1}{3} + \frac{k}{2} \Rightarrow k = \frac{2}{3}$, $f(x) = x + \frac{2}{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 71, 73-75; J-II 74-76, 87, 94-96; J-M 16, 32, 50, 52, 74, 78; J-A 44; C-II 34-36 | Question Bank: 26, 62-63, 67, 87

Also used in: Sheet: J-M 43; C-II 31, 38

Definite Integral as a Limit of Sum

Question 30

How to convert an expression given in sigma form to integral form?

Model Answer

$$\lim_{n\to\infty} \frac{1}{n} \sum_{r=0}^{n-1} f\left(\frac{r}{n}\right) = \int_0^1 f(x)\,dx$$

Write the general term as $\frac{1}{n} f\left(\frac{r}{n}\right)$ (take out the highest power of n); replace $\lim \sum$ by $\int$, $\frac{r}{n}$ by x, $\frac{1}{n}$ by dx.

Ex: $\lim_{n\to\infty} \sum_{r=1}^{n} \frac{n}{n^2+r^2} = \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 77-80; J-II 77, 102; J-M 6, 12, 22, 38, 61, 63, 73; J-A 3, 29; C-I 20-22, 29, 31; C-II 39, 41 | DPP: D4(10-13, 15-16) | Question Bank: 52, 75

Also used in: Sheet: J-II 78

Question 31

How to compute the upper and lower limits while converting?

Model Answer

Lower limit $= \lim \frac{r}{n}$ at the first value of r, upper limit $= \lim \frac{r}{n}$ at the last value: $r = 1$ to n gives 0 to 1; $r = 1$ to pn gives 0 to p.

Ex: $\lim_{n\to\infty} \sum_{r=1}^{2n} \frac{1}{n+r} = \int_0^2 \frac{dx}{1+x} = \ln 3$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 32

What to do if the expression is given in the form of a continued product?

Model Answer

Put the limit $= L$ and take ln: the product becomes a sum $\frac{1}{n} \sum \ln f\left(\frac{r}{n}\right)$, so $\ln L = \int_0^1 \ln f(x)\,dx$.

Ex: $\lim_{n\to\infty} \left(\frac{(n+1)(n+2)\cdots(2n)}{n^n}\right)^{1/n} = e^{\int_0^1 \ln(1+x)dx} = \frac{4}{e}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 81; J-II 78; J-M 26; J-A 22, 28 | DPP: D4(14)

Also used in: none

Question 33

For a finite n, how to check the order among $\frac{1}{n} \sum_{r=0}^{n-1} f\left(\frac{r}{n}\right)$, $\int_0^1 f(x)\,dx$ and $\frac{1}{n} \sum_{r=1}^{n} f\left(\frac{r}{n}\right)$?

Model Answer

These are left rectangles Q, area P, right rectangles R. f increasing: $Q < P < R$; f decreasing: $R < P < Q$.

Ex: $f = x^2$, $n = 2$: $\frac{1}{8} < \frac{1}{3} < \frac{5}{8}$.

Grade - Low

Practice map, main concept: Sheet: J-II 79, 101, 105

Also used in: none

Definite Integration by Estimation

Question 34

How to estimate a definite integral with the help of monotonicity?

Model Answer

f increasing on $[a,b]$: $(b-a)f(a) < \int_a^b f(x)\,dx < (b-a)f(b)$ (reverse for decreasing)

Non-monotonic: $(b-a)\cdot\min f \le \int_a^b f \le (b-a)\cdot\max f$ (find max/min by derivative).

Ex: e^{x^2} increases on $[0,1]$, so $1 < \int_0^1 e^{x^2}dx < e$.

Grade - Medium

Practice map, main concept: Sheet: J-II 80; J-M 66; C-II 33

Also used in: none

Question 35

How to estimate a definite integral with the help of concavity?

Model Answer

Concave up: $(b-a)\,f\left(\dfrac{a+b}{2}\right) < \int_a^b f(x)\,dx < (b-a)\,\dfrac{f(a)+f(b)}{2}$ (reverse for concave down)

The chord (trapezium) lies above a concave-up curve, the tangent at the midpoint lies below it.

Ex: e^x on $[0,1]$: $e^{1/2} \approx 1.649 < e-1 \approx 1.718 < \dfrac{1+e}{2} \approx 1.859$.

Grade - Low

Practice map, main concept: Sheet: J-II 93

Also used in: none

Question 36

How to estimate a definite integral by splitting the interval into two or more parts?

Model Answer

$$\int_a^b f(x)\,dx \approx \frac{b-a}{4}\left[f(a)+2f\left(\frac{a+b}{2}\right)+f(b)\right]$$

Apply the bounds of 34 / 35 on each part and add; more parts give a sharper estimate (the formula above uses the midpoint).

Ex: $\int_0^{\pi/2}\sin x\,dx \approx \dfrac{\pi}{8}\left(0+\sqrt{2}+1\right) \approx 0.948$ (exact value 1).

Grade - Low

Practice map, main concept: Sheet: J-II 91

Also used in: none

Question 37

How to estimate a definite integral by assumption of functions?

Model Answer

$f_1(x) \le f(x) \le f_2(x)$ on $[a,b] \Rightarrow \int_a^b f_1 \le \int_a^b f \le \int_a^b f_2$; $\left|\int_a^b f\right| \le \int_a^b |f|$

Compare with an easy function: on $(0,1)$, $x^3 < x^2$; $\sin x < x$; $1+x < e^x$; $\sqrt{1-x^2} \ge 1-x^2$, etc.

Ex: on $(0,1)$, $2^{x^3} < 2^{x^2}$, so $\int_0^1 2^{x^3}dx < \int_0^1 2^{x^2}dx$.

Grade - Medium

Practice map, main concept: Sheet: J-I 82-83; J-A 17, 23, 33, 43 | DPP: D4(17-18)

Also used in: Sheet: J-II 80

Walli's Theorem

Question 38

Write Walli's Theorem and use it in some basic examples.

Model Answer

$$\int_0^{\pi/2} \sin^n x \cos^m x \, dx = \frac{[(n-1)(n-3)\cdots][(m-1)(m-3)\cdots]}{(m+n)(m+n-2)\cdots} \cdot K$$

Each product stops at 1 or 2; $K = \frac{\pi}{2}$ if m and n are both even, otherwise $K = 1$. With $m = 0$: $\int_0^{\pi/2} \sin^n x \, dx$.

Ex: $\int_0^{\pi/2} \sin^4 x \cos^6 x \, dx = \frac{(3 \cdot 1)(5 \cdot 3 \cdot 1)}{10 \cdot 8 \cdot 6 \cdot 4 \cdot 2} \cdot \frac{\pi}{2} = \frac{3\pi}{512}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 81 | DPP: D4(19-20, 22-23)

Also used in: Sheet: J-A 5 | DPP: D4(21)

Integration by Reduction

Question 39

What is the basic theme to solve integration by reduction questions?

Model Answer

Use parts (often with 1 as the second function, or split one factor off) to relate I_n to I_{n-1} or I_{n-2}; in definite form the boundary term $[uv]_a^b$ often vanishes. Then use the relation repeatedly.

$\int_1^e (\ln x)^n dx: \; I_n = e - nI_{n-1}$; $\int_0^{\pi/4} \tan^n x \, dx: \; I_n + I_{n-2} = \frac{1}{n-1}$

Ex: $\int_1^e (\ln x)^2 dx = e - 2\int_1^e \ln x \, dx = e - 2$.

Grade - Medium

Practice map, main concept: Sheet: J-I 84-85; J-II 82-84; J-M 31, 40, 69, 76; C-I 28; C-II 28-30 | Question Bank: 32, 41, 43, 73, 86

Also used in: Sheet: J-II 106